BANKING
50%
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INTERNET BANKING SYSTEM

Submitted By :

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Guided By :

Ms. Nidhi Naik

PROJECT PROFILE

Field Name	Description
Project Name	Internet Banking System
Objective	To provide a secure, 24/7 web platform for instant fund transfers and account management.
Frontend Tool	HTML, CSS, JavaScript, Bootstrap 4
Backend Tool	PHP (Server-side Scripting)
Database	MySQL (MariaDB)
Guide Name	Ms. Nidhi Naik
Group Members	Parmar Pratik , Patel Sujal , Patel Raj

EXISTING SYSTEM

- **Manual Data Entry:** Records are maintained in physical registers or basic spreadsheets, leading to high chances of human error.
- **Limited Accessibility:** Customers can only perform transactions during bank hours, causing inconvenience during emergencies.
- **Physical Presence Required:** For every small task like a balance enquiry or fund transfer, the customer must visit the branch and stand in long queues.
- **Security Risks:** Manual documents are prone to theft, fire, or being misplaced, and lack the high-level encryption found in digital systems.
- **Slow Processing**: Fund transfers between different branches or banks can take several business days to reflect.

NEED FOR NEW SYSTEM

- **24/7 Online Access**: Provides the freedom to manage accounts and transfer funds at any time (24 hours a day, 7 days a week).

- **Instant Transactions:** Funds are transferred and balances are updated in real-time using automated database triggers.

- **Enhanced Security:** Implementation of double-layered encryption (SHA1 and MD5) protects user passwords and sensitive transaction data.

- **User-Friendly Interface:** A responsive web dashboard allows users to access banking services from any device (Mobile, Tablet, or PC).

- **Paperless Environment:** Reduces the need for physical paperwork, making the banking process eco-friendly and more organized.

SYSTEM REQUIREMENTS

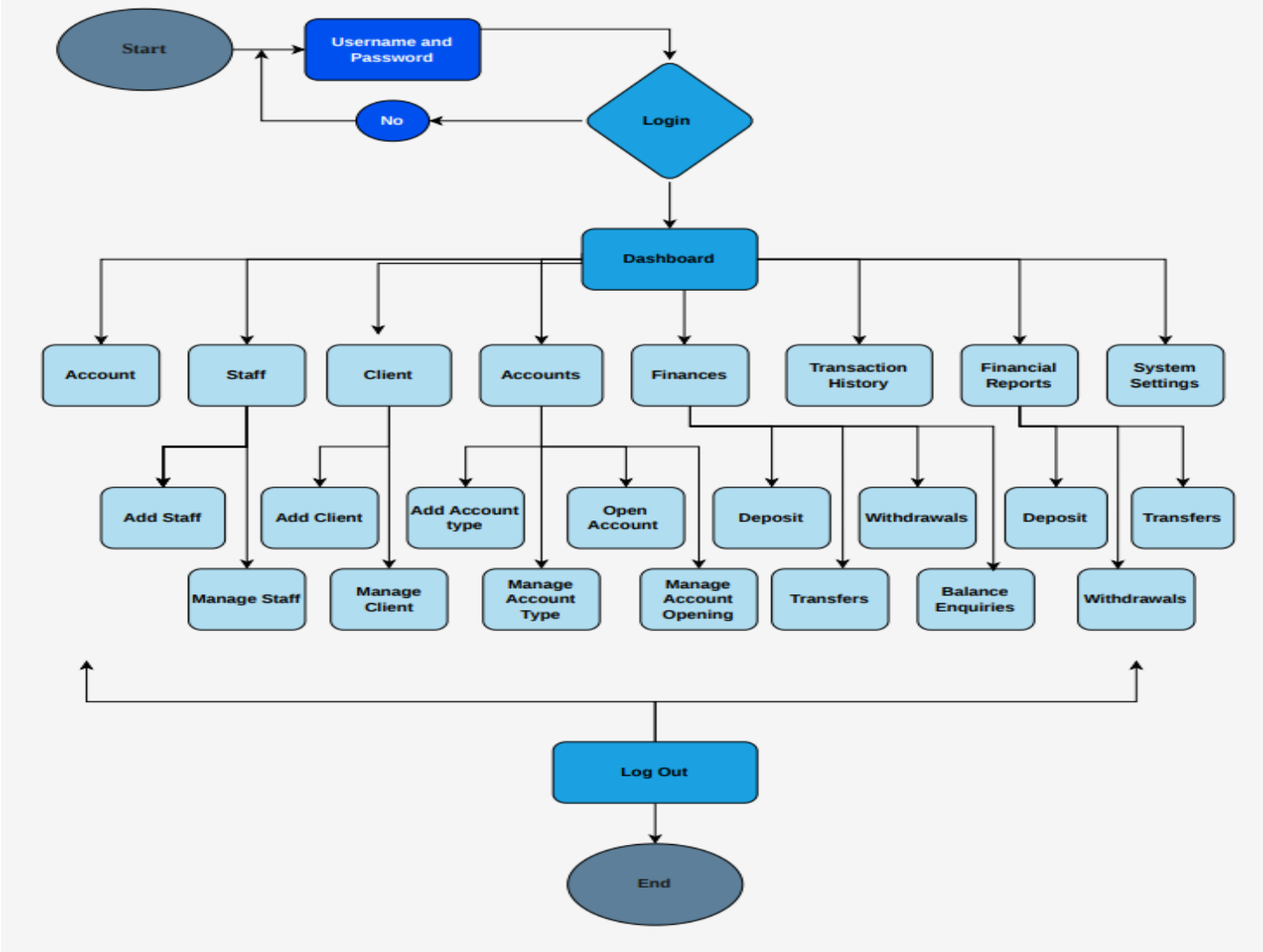
1. **Hardware Requirements:**

- **Processor:** Intel Core i3 or higher.
- **RAM:** Minimum 4 GB (8 GB Recommended).
- **Hard Disk:** 500 MB free space for project files and database.
- **Input Devices:** Standard Keyboard and Mouse.
- **Monitor:** 1366 x 768 resolution or higher.

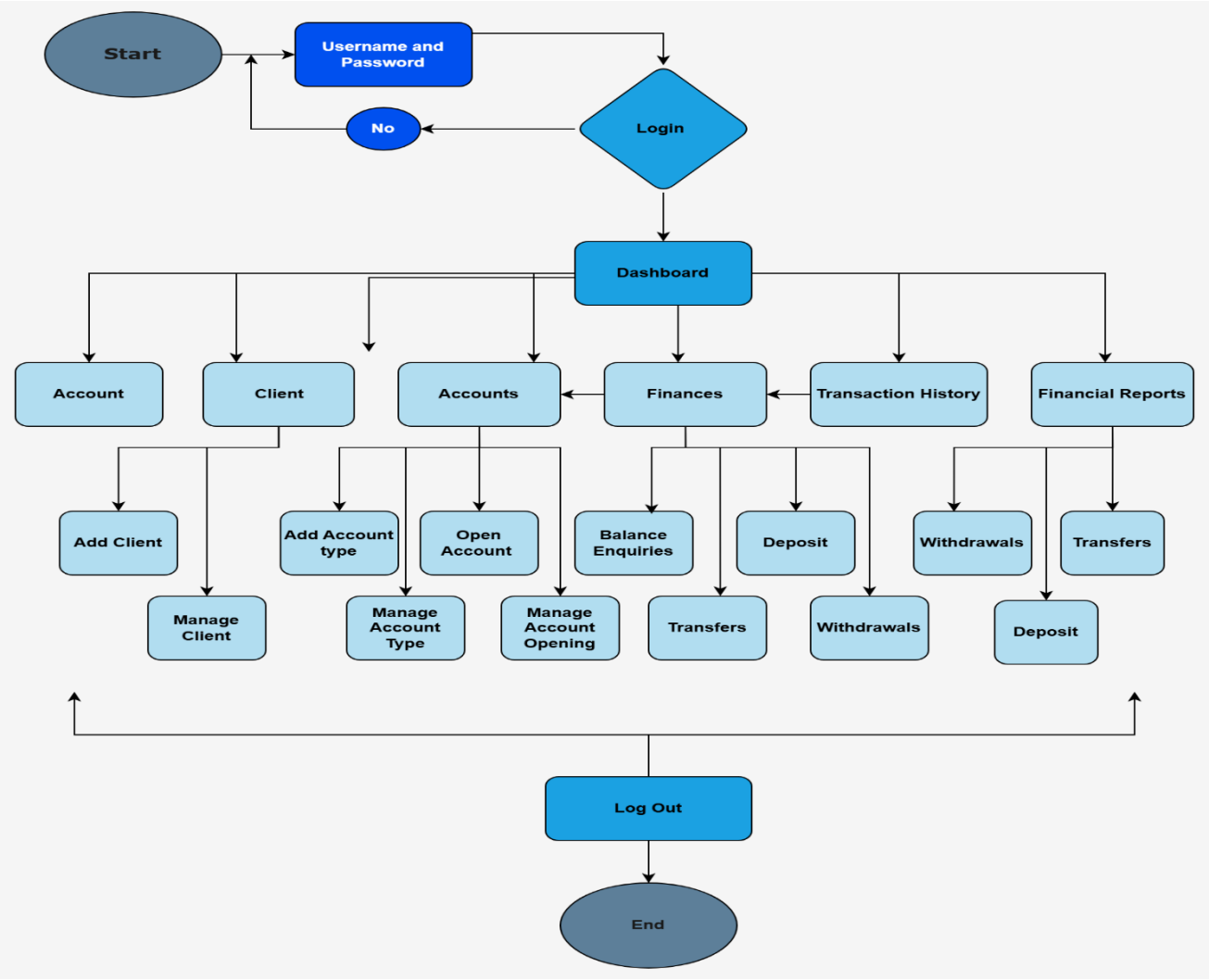
2. Software Requirements:

- **Operating System:** Windows 10/11 or Linux.
- **Web Server:** XAMPP Control Panel (Apache Server).
- **Language:** PHP 7.4 or 8.x.
- **Database:** MySQL / MariaDB.
- **Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge.
- **Text Editor:** VS Code, Sublime Text, or Notepad++.

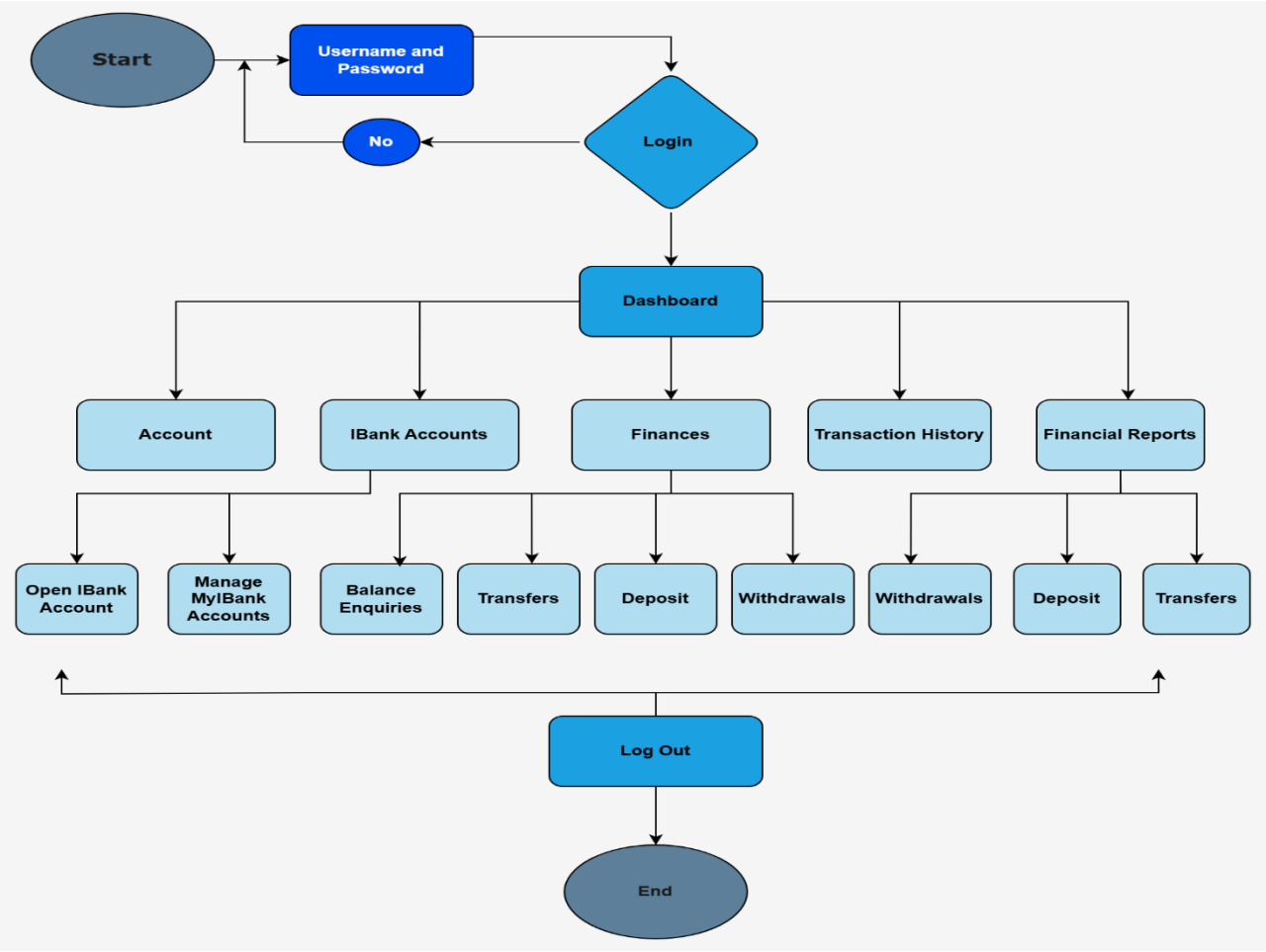
ADMIN FLOWCHART



STAFF FLOWCHART



CLIENT FLOWCHART


Start
Username and Password
No
Login
Dashboard
Account
IBank Accounts
Finances
Transaction History
Financial Reports
Open IBank Account
Manage MyIBank Accounts
Balance Enquiries
Transfers
Deposit
Withdrawals
Withdrawals
Deposit
Transfers
Log Out
End

DATABASE TABLE OVERVIEW

Sr. No.	Table Name	Purpose / Description
1	**ib_admin**	Isme System Administrator ke login details aur rights store hote hain jo poore portal ko control karta hai.
2	**ib_staff**	Bank employees (Staff) ki details aur unka authentication data manage karne ke liye use hoti hai.
3	**ib_clients**	Customers ki personal information jaise Name, Email, aur Contact details ko track karne ke liye hai.
4	**ib_acc_types**	Bank mein alag-alag types ke accounts (Saving, Current, Fixed Deposit) aur unke details define karti hai.
5	**ib_bankaccounts**	Yeh main table hai jo Client ID ko unke Account Number aur current balance ke saath connect karti hai.
6	**ib_notifications**	System alerts, news, aur transaction updates ko user ke dashboard par dikhane ke liye data store karti hai.
7	**ib_systemsettings**	Website ka name, logo path, aur system-wide configurations ko customize karne ke liye use hoti hai.
8	**ib_transactions**	Har ek paise ka hisaab (Deposit, Withdraw, Transfer) isme transaction code aur timestamp ke saath save hota hai.

ADMIN TABLE

Column	Data Type	Description
id	int(11)	Unique ID for the Administrator.
name	varchar(255)	Full name of the Admin.
email	varchar(255)	Official login email.
password	varchar(255)	Encrypted password string.

STAFF TABLE

Column	Data Type	Description
s_id	int(11)	Unique Staff ID.
s_name	varchar(255)	Name of the staff member.
s_email	varchar(255)	Staff login email.
s_pwd	varchar(255)	Encrypted staff password.
s_status	varchar(255)	Current status (Active/Blocked).

CLIENT TABLE

Column	Data Type	Description
c_id	int(11)	Unique Client ID.
c_name	varchar(255)	Full name of the account holder.
c_email	varchar(255)	Registered email for updates.
c_phone	varchar(255)	Mobile number for SMS alerts.
c_pwd	varchar(255)	Password for customer portal.
c_address	text	Permanent residential address.
c_profile	varchar(255)	Profile picture file name.

ACCOUNT TYPES TABLE

Column	Data Type	Description
acct_id	int(11)	Unique ID for account category.
acct_type	varchar(255)	Name (Saving, Current, etc.).
acct_rate	decimal(10,2)	Yearly interest rate percentage.

BANK ACCOUNTS TABLE

Column	Data Type	Description
acc_id	int(11)	Unique Internal ID.
acc_no	varchar(255)	Unique Bank Account Number.
acc_type	varchar(255)	Category of the account.
acc_balance	decimal(65,2)	Total current available balance.
client_id	int(11)	ID linking to ib_clients table.
acc_status	varchar(255)	Current status of the account.

NOTIFICATIONS TABLE

Column	Data Type	Description
n_id	int(11)	Notification ID.
n_content	text	Message body for the user.
n_date	timestamp	Time when notification was sent.

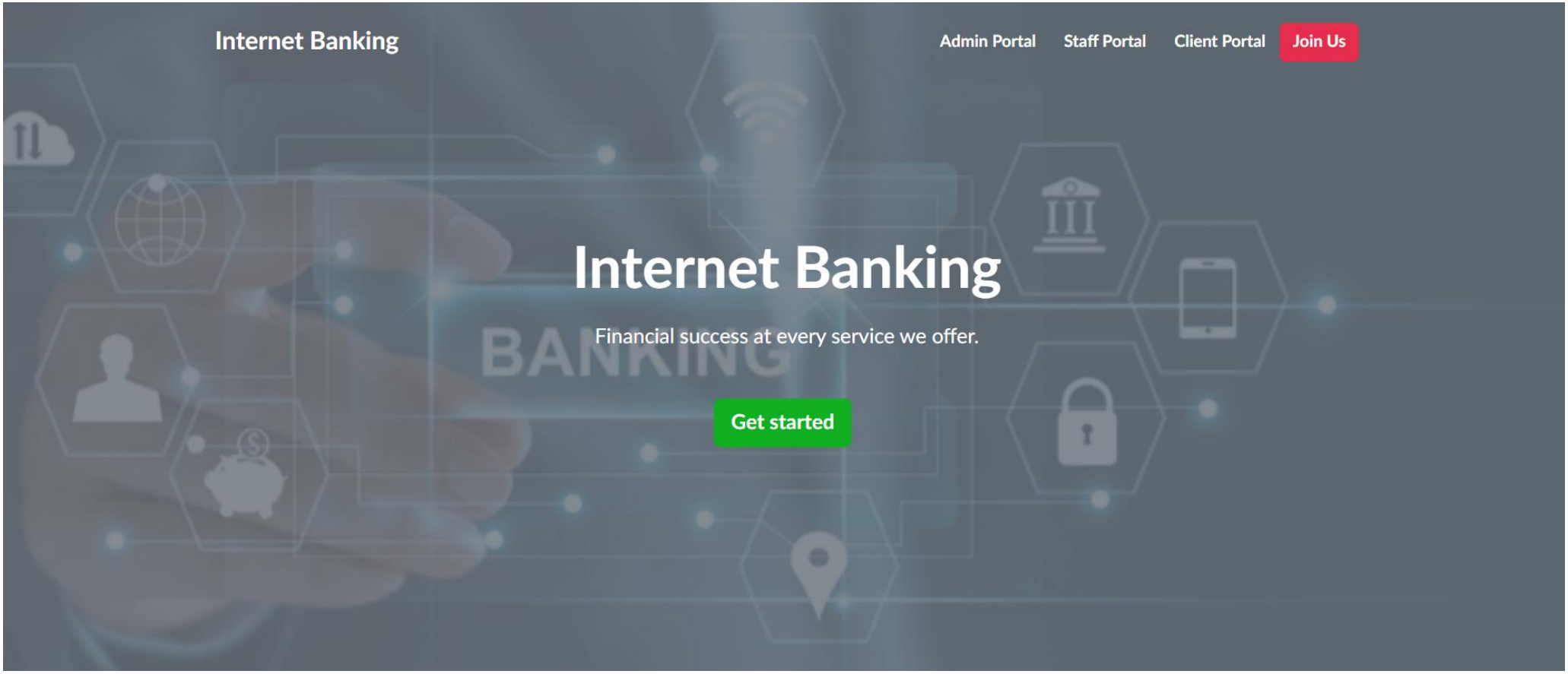
SYSTEM SETTINGS TABLE

Column	Data Type	Description
sys_id	int(11)	Setting ID.
sys_name	varchar(255)	Bank name for branding.
sys_logo	varchar(255)	Path to the bank logo file.

TRANSACTIONS TABLE

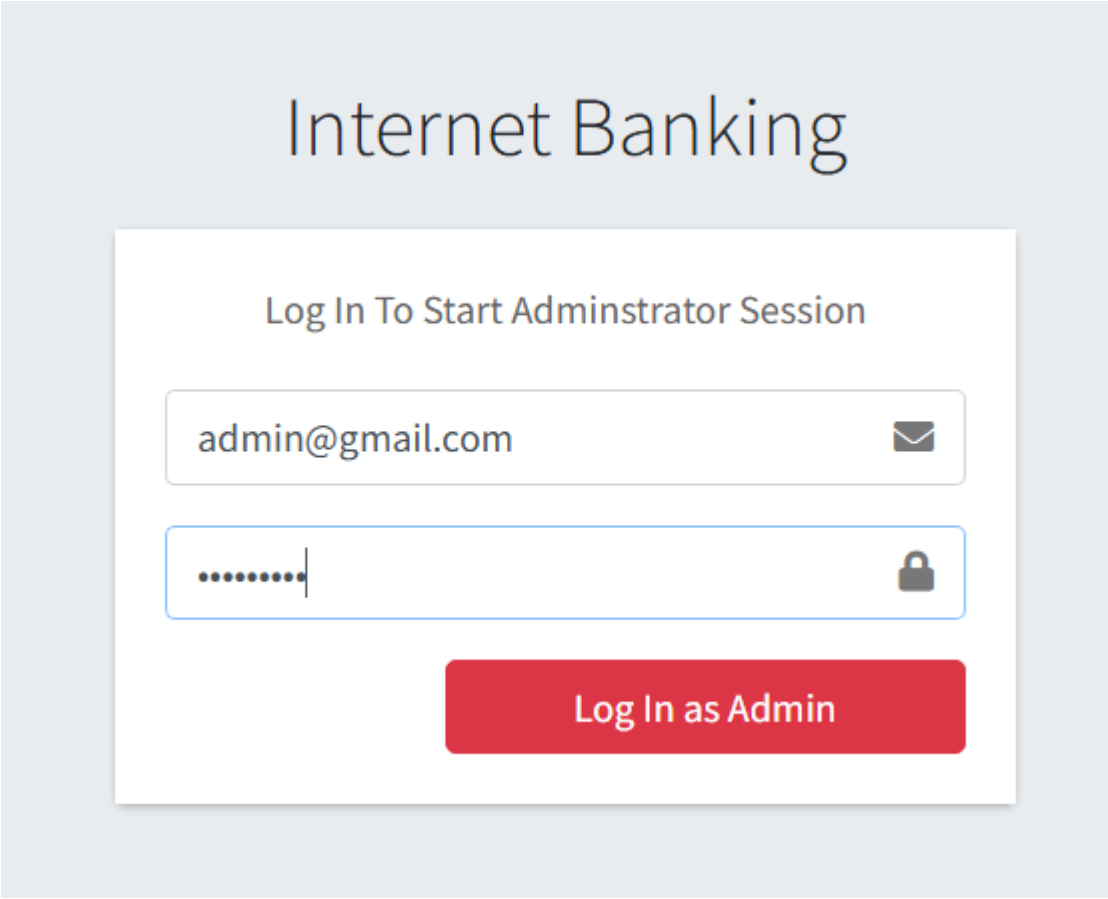
Column	Data Type	Description
tr_id	int(11)	Unique Transaction ID.
tr_code	varchar(255)	Unique Tracking/Reference code.
tr_type	varchar(255)	Type (Credit or Debit).
tr_amount	decimal(65,2)	Exact amount transacted.
tr_acc_no	varchar(255)	Account number involved.
tr_date	timestamp	Automated date and time.

HOME PAGE



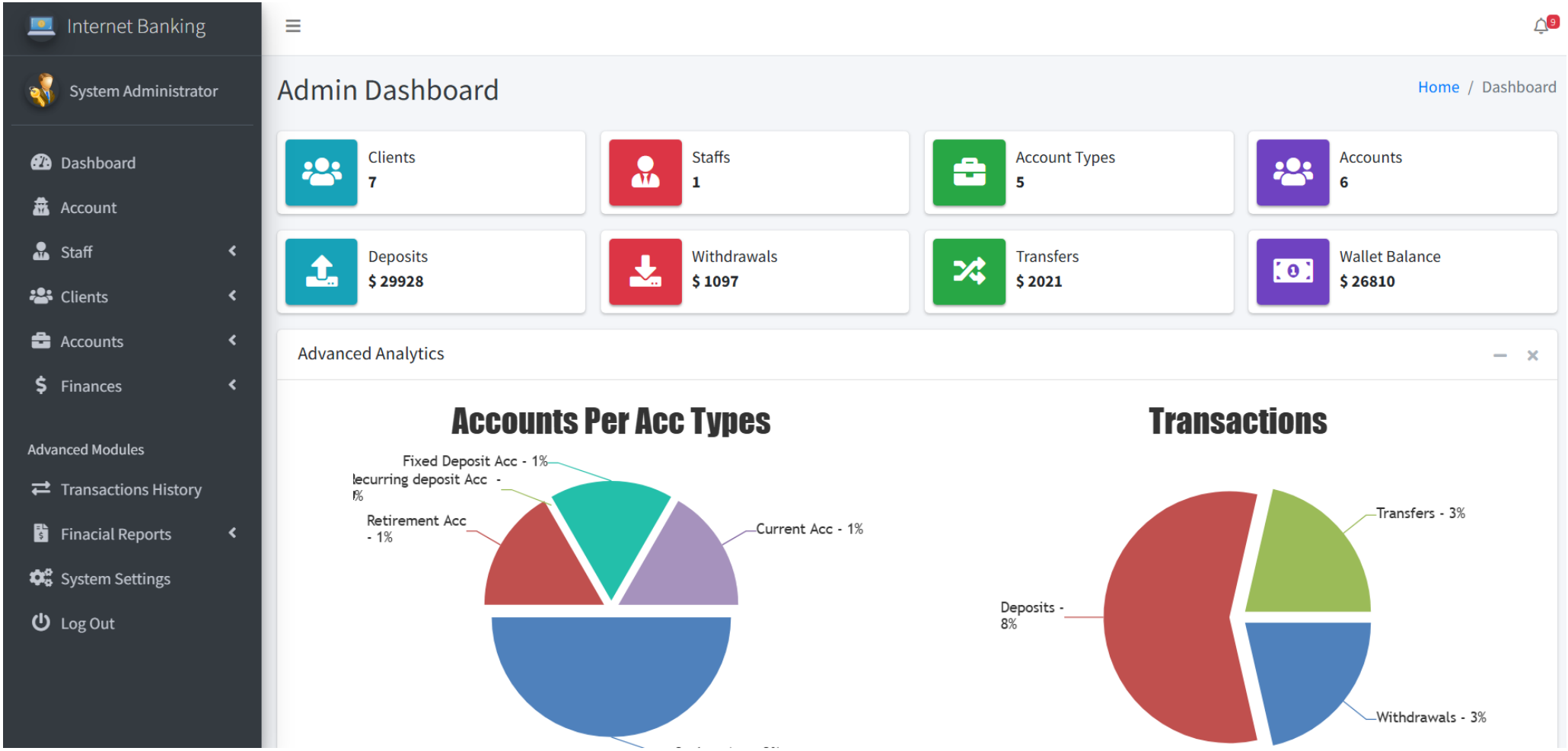
This is the main index page of the Internet Banking System. It serves as the landing page for all visitors, providing a brief overview of banking services and quick links for User and Staff login.

ADMIN AUTHENTICATION



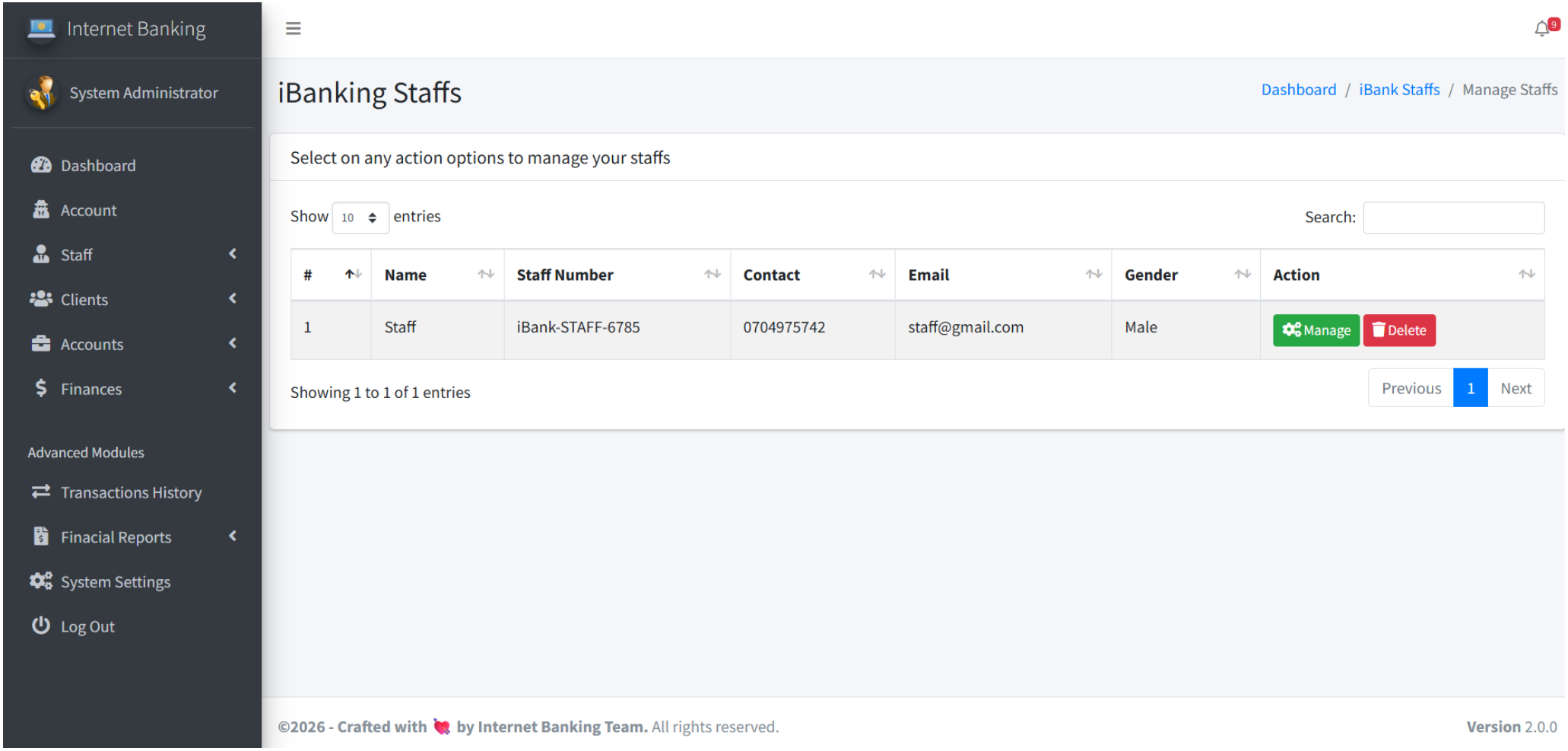
The secure login portal for the Administrator. It uses session management to ensure that only the authorized person can access the management modules.

ADMIN DASHBOARD



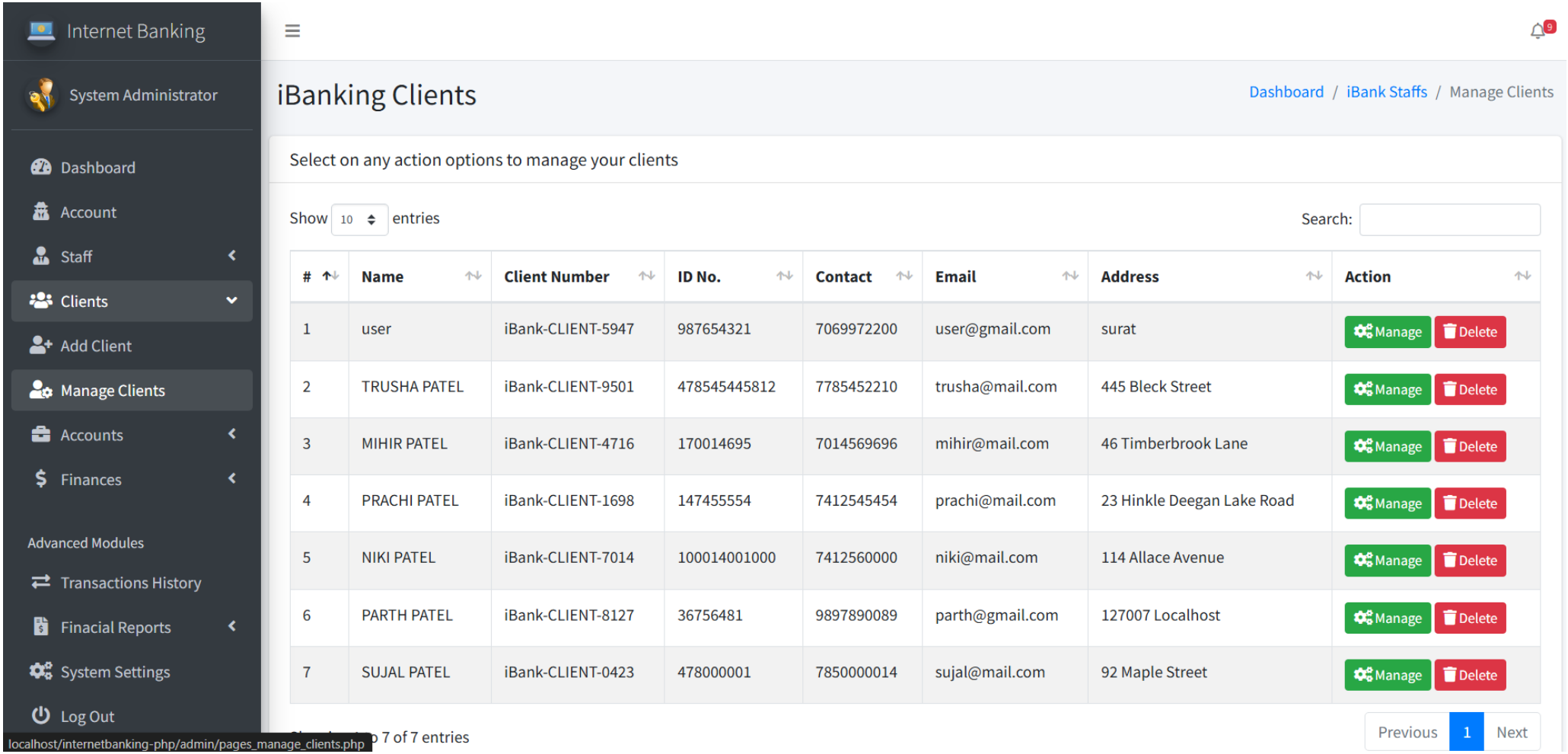
The Dashboard provides real-time analytics. The Admin can track total registered clients, active staff members, and system-wide account activities from a single interface.

MANAGE STAFF (ADMIN SIDE)



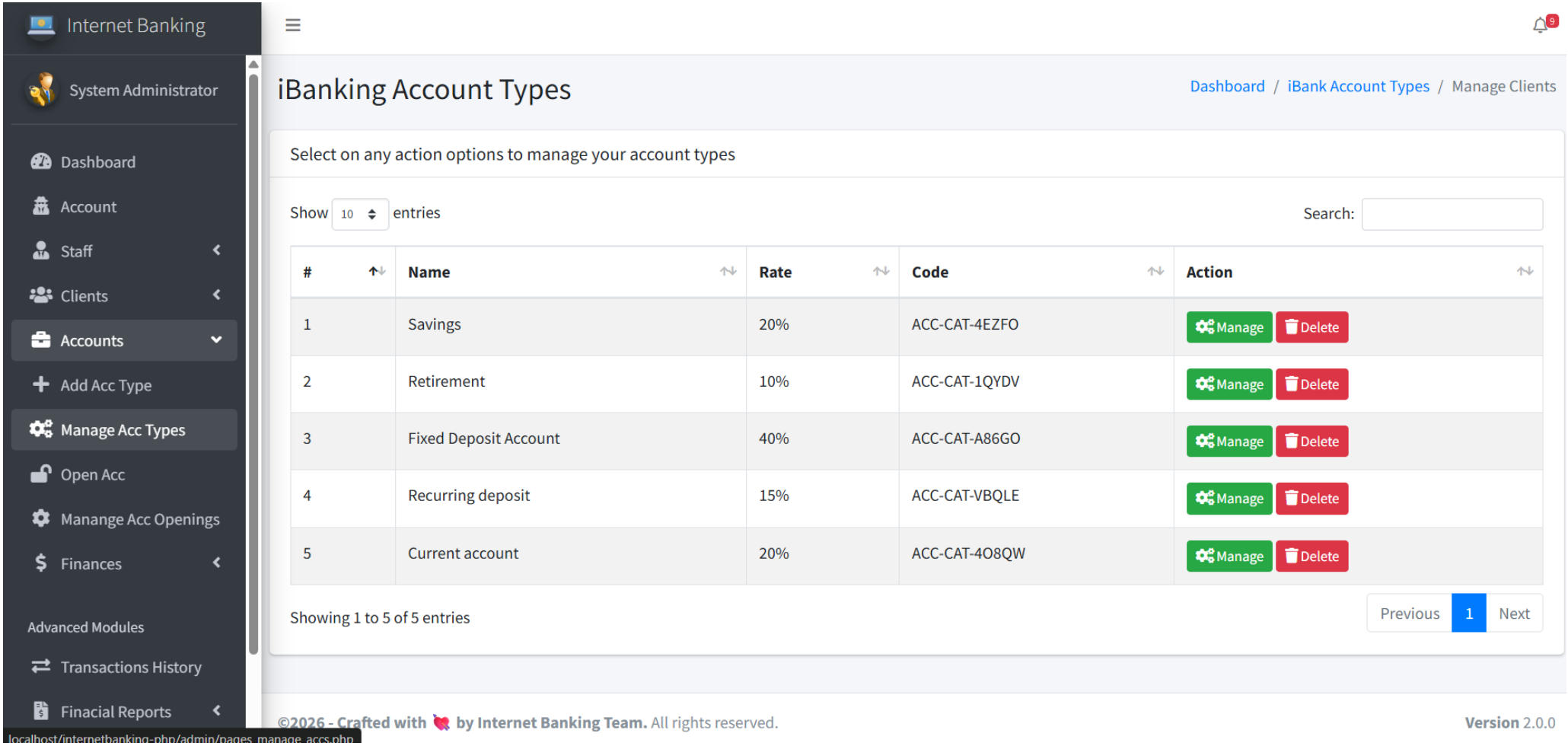
Admin has the authority to hire/register new bank staff and manage their profiles.

MANAGE CLIENT (ADMIN SIDE)



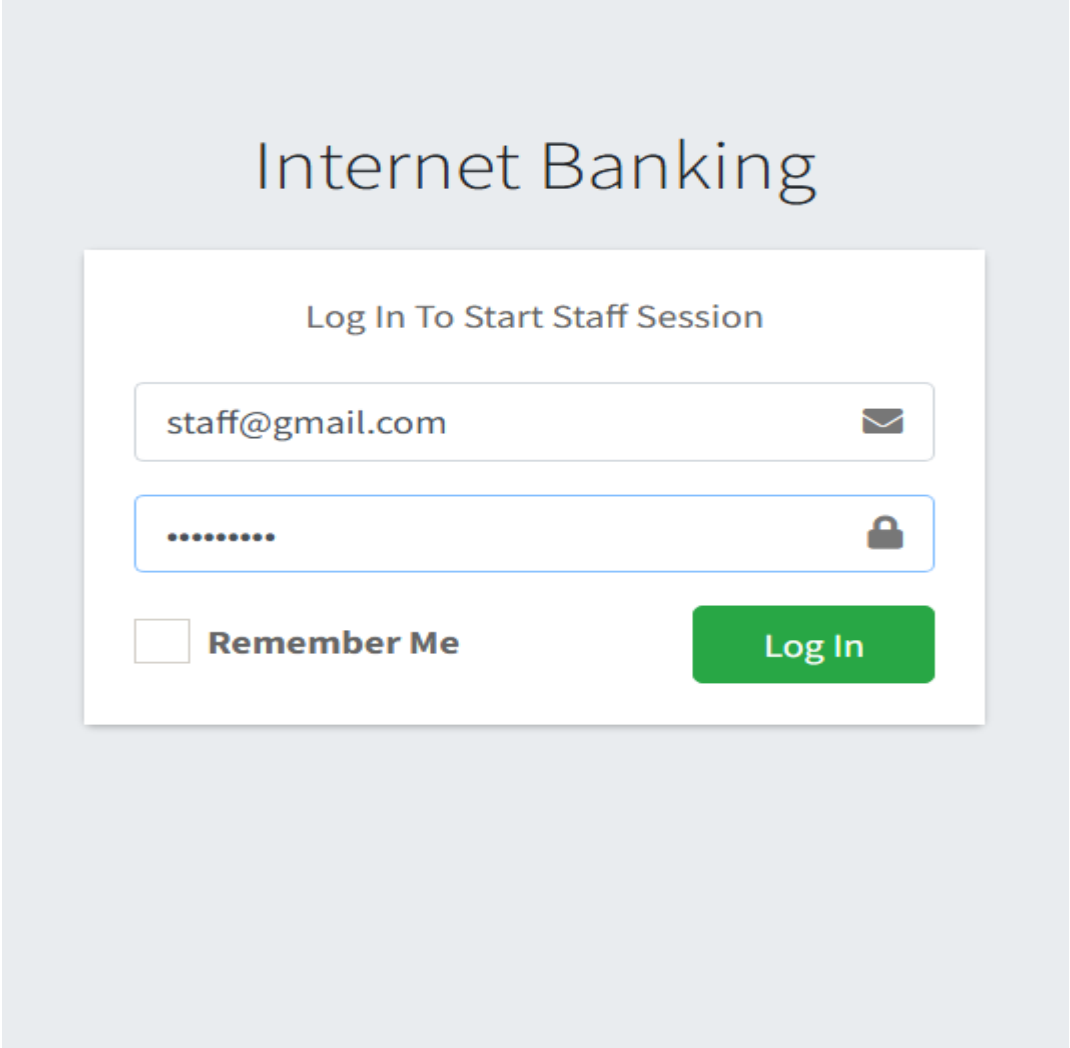
Admin has the authority to hire/register new bank client and manage their profiles.

ACCOUNT TYPE (ADMIN SIDE)



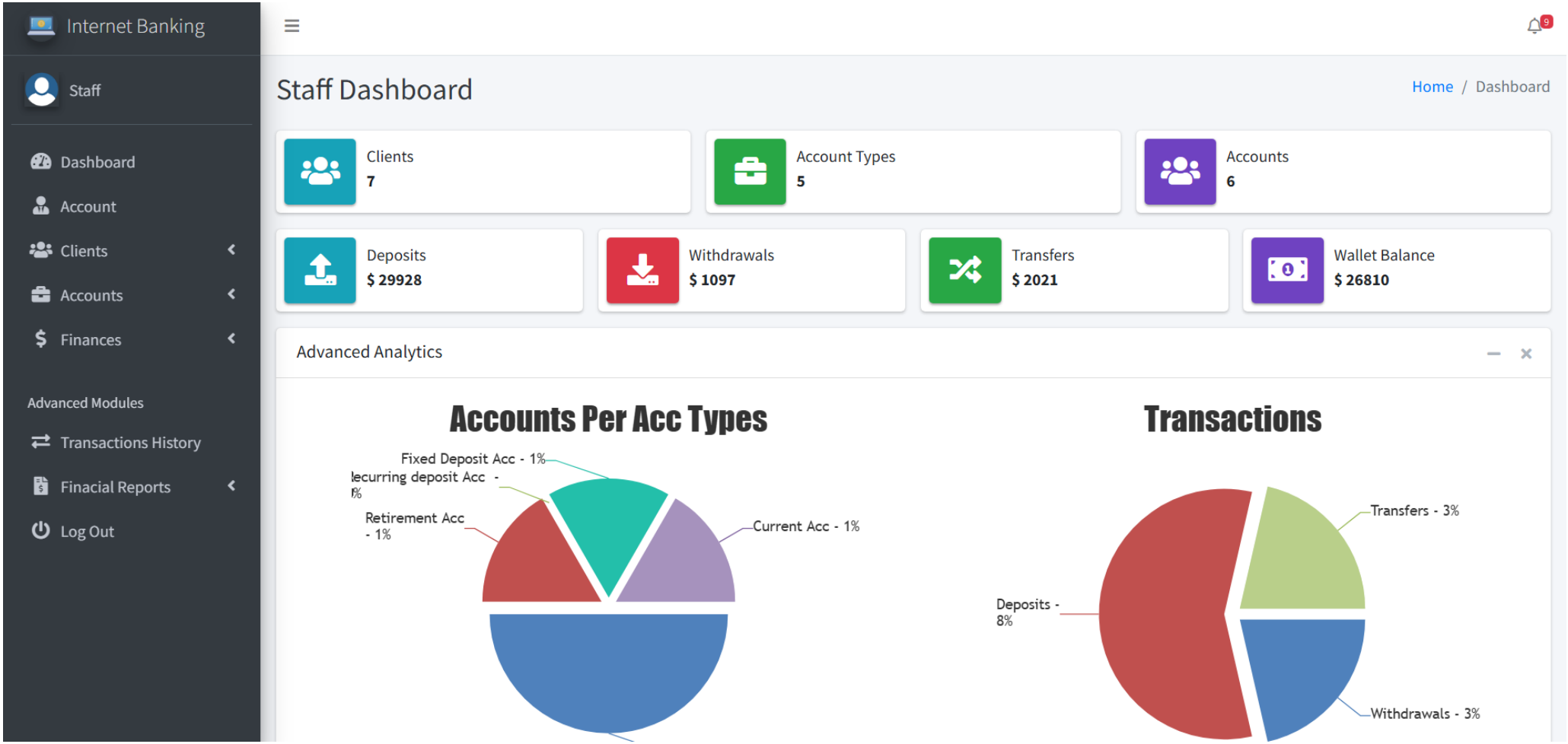
This module allows the admin to define various bank account categories like Savings or Current accounts along with their respective interest rates.

AUTHENTICATION (STAFF SIDE)



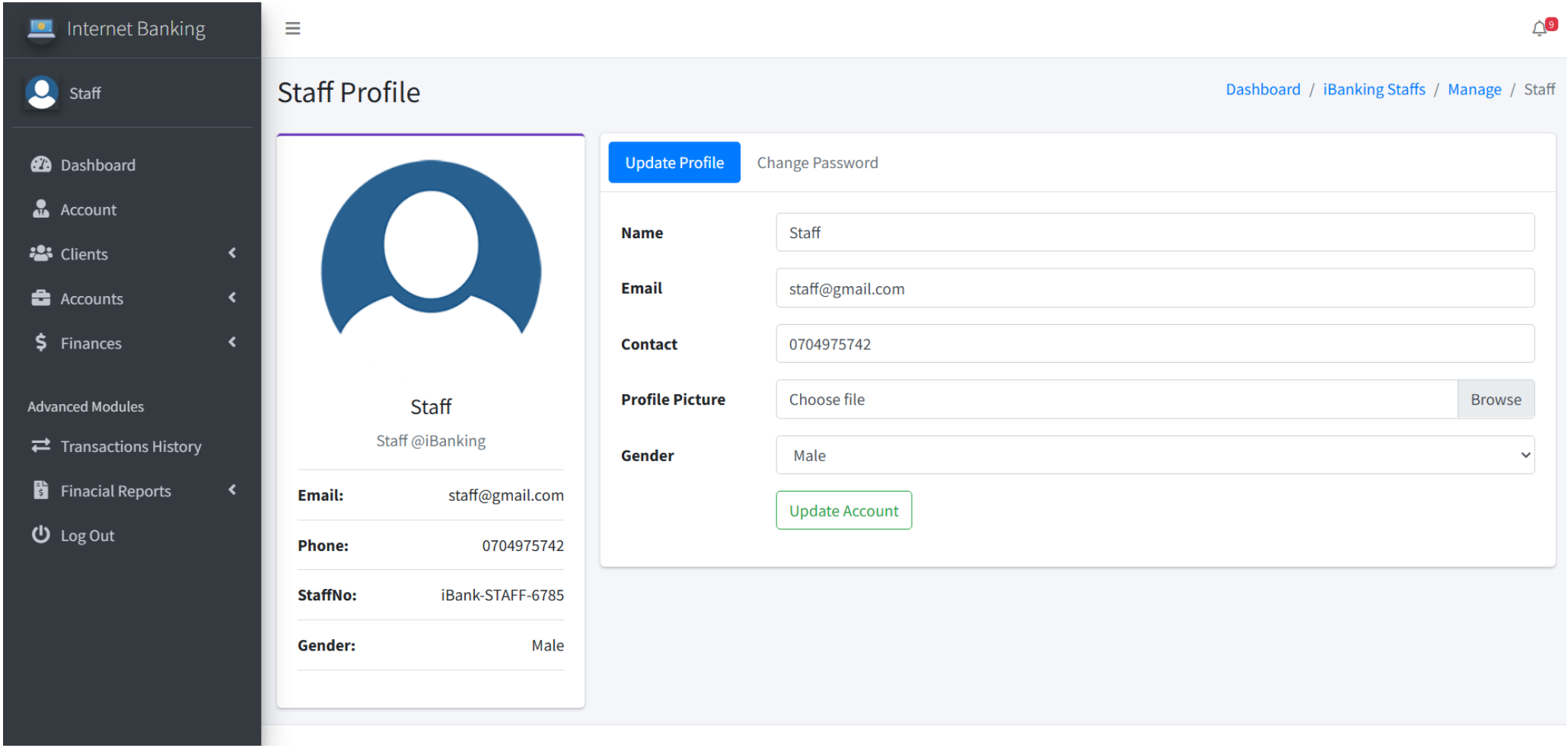
This is the secure login interface specifically designed for bank employees (Staff). It ensures that staff members can only access operational modules assigned to them by the Administrator.

DASHBOARD (STAFF SIDE)



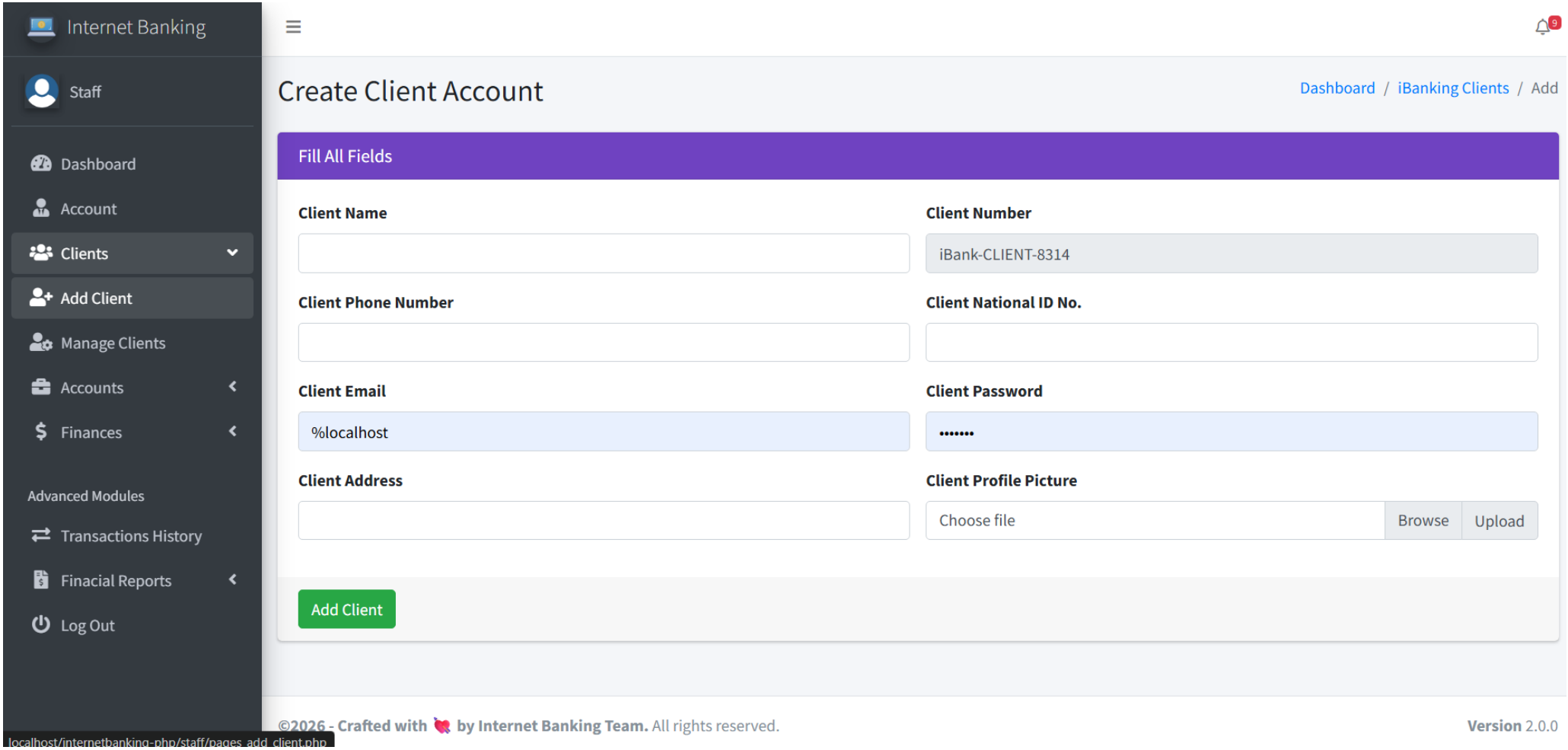
Upon a successful login, the staff is redirected to this dashboard. It provides a quick summary of the work handled, including the total number of clients registered, active bank accounts, and transactions processed by the staff.

PROFILE MANAGEMENT (STAFF SIDE)



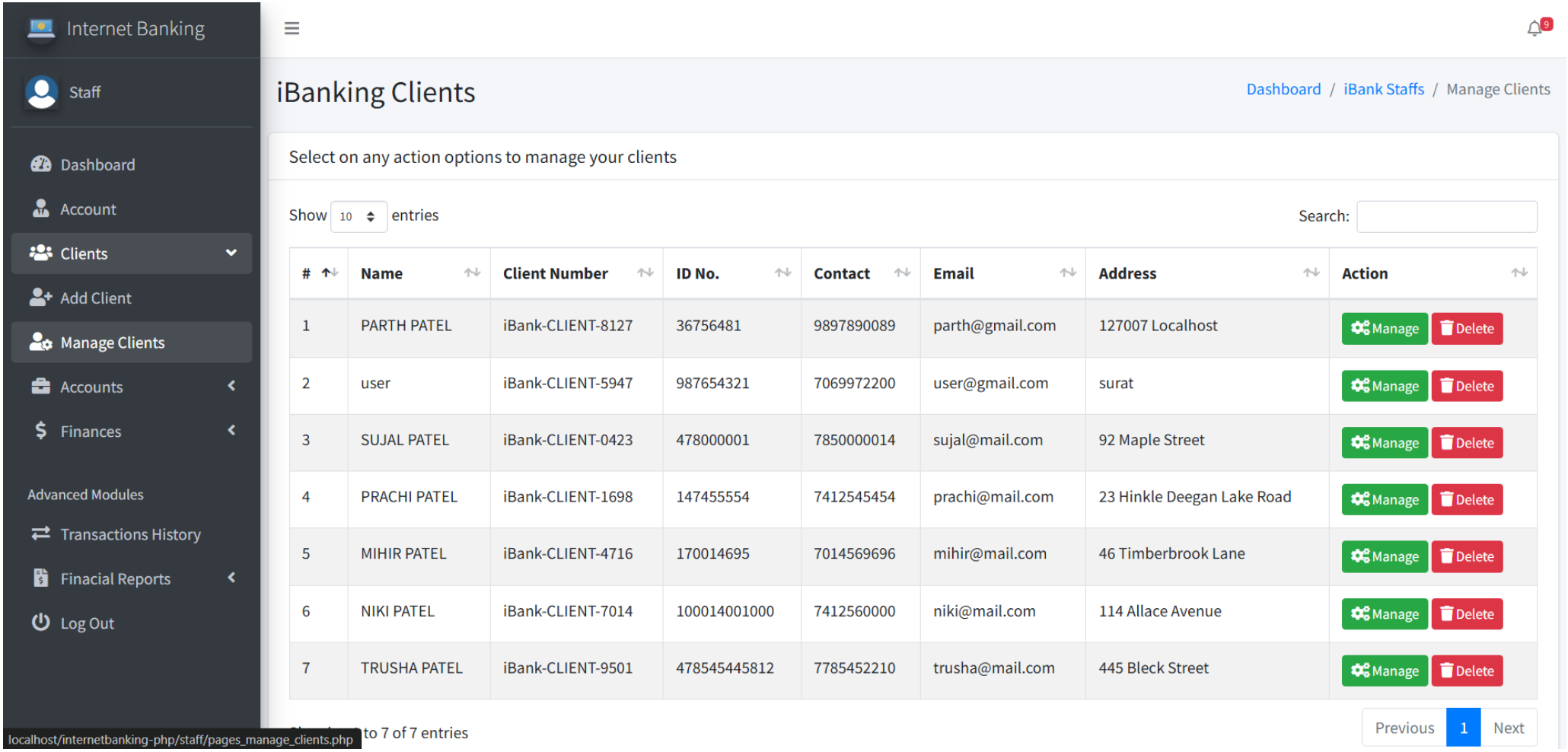
This module allows the authorized user (Admin or Staff) to view and update their personal profile details within the system. It ensures that the bank's internal records are always accurate and reflect the current contact information of the personnel.

CLIENT REGISTRATION (STAFF SIDE)



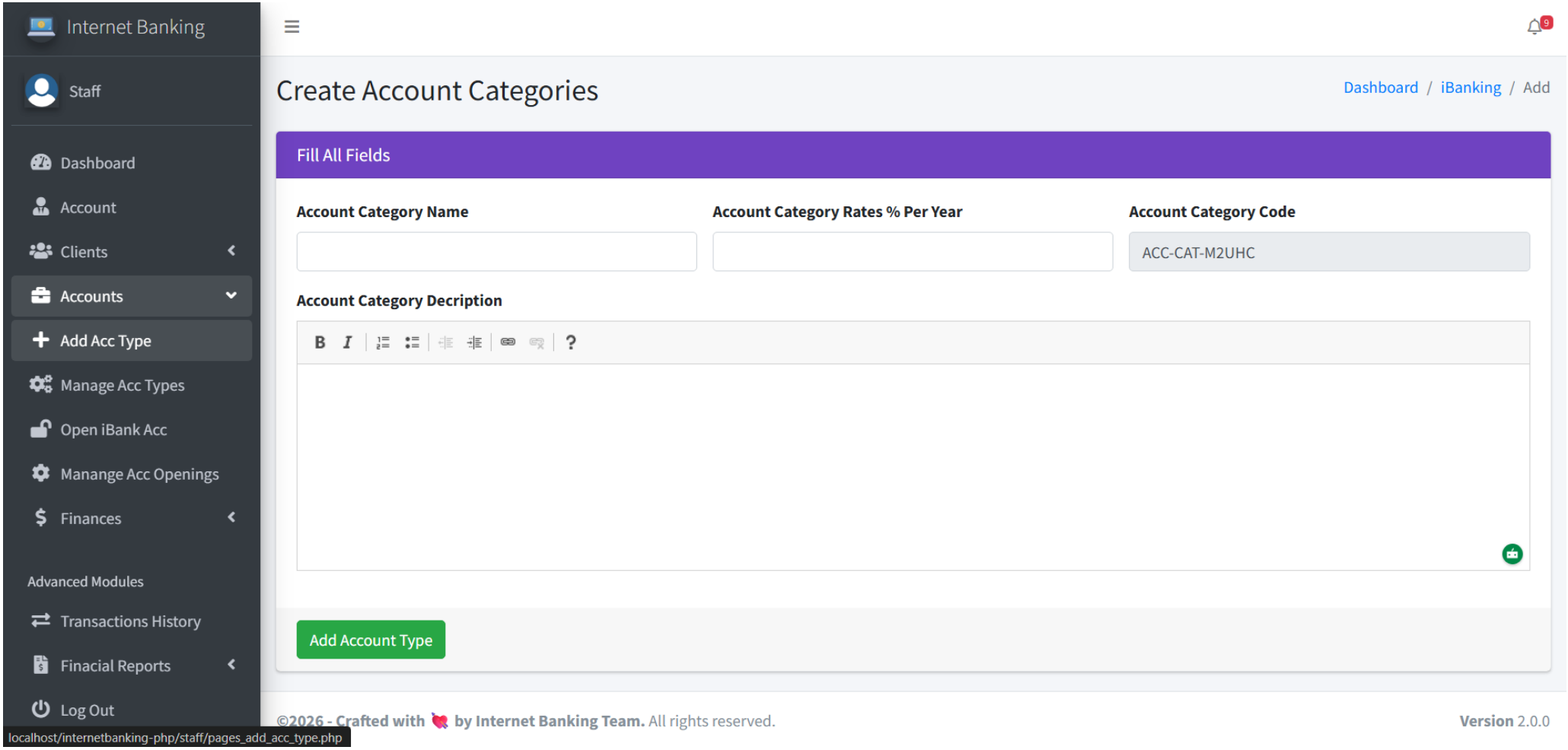
This form-based module is used by the staff to register new customers into the system. It captures essential details such as the client's name, contact information, and initial account preferences.

MANAGE CLIENT LIST (STAFF SIDE)



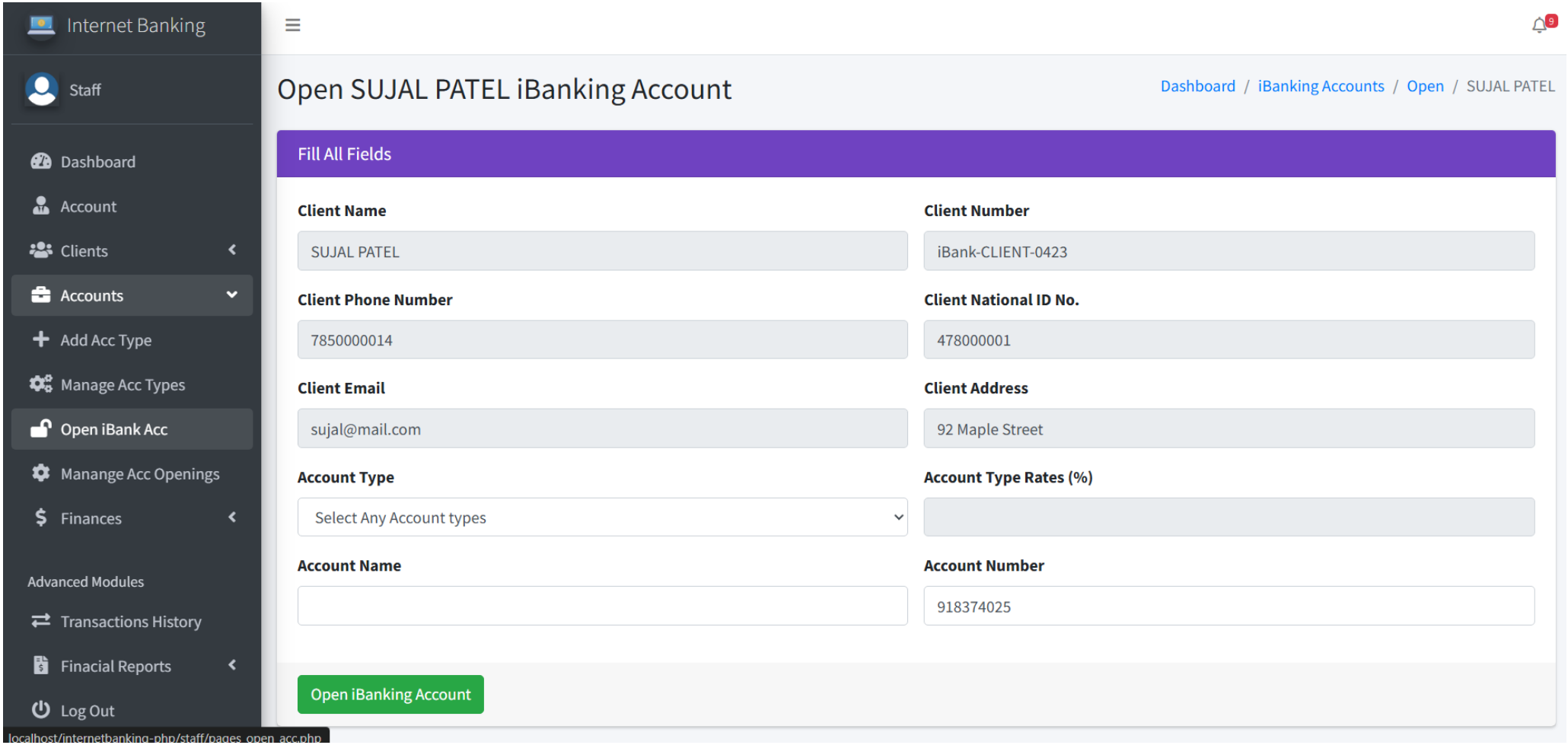
This screen displays a comprehensive list of all clients registered under the staff's branch. It allows the staff to view, update, or verify customer profiles to ensure data accuracy.

ACCOUNT TYPE REGISTRATION (STAFF SIDE)



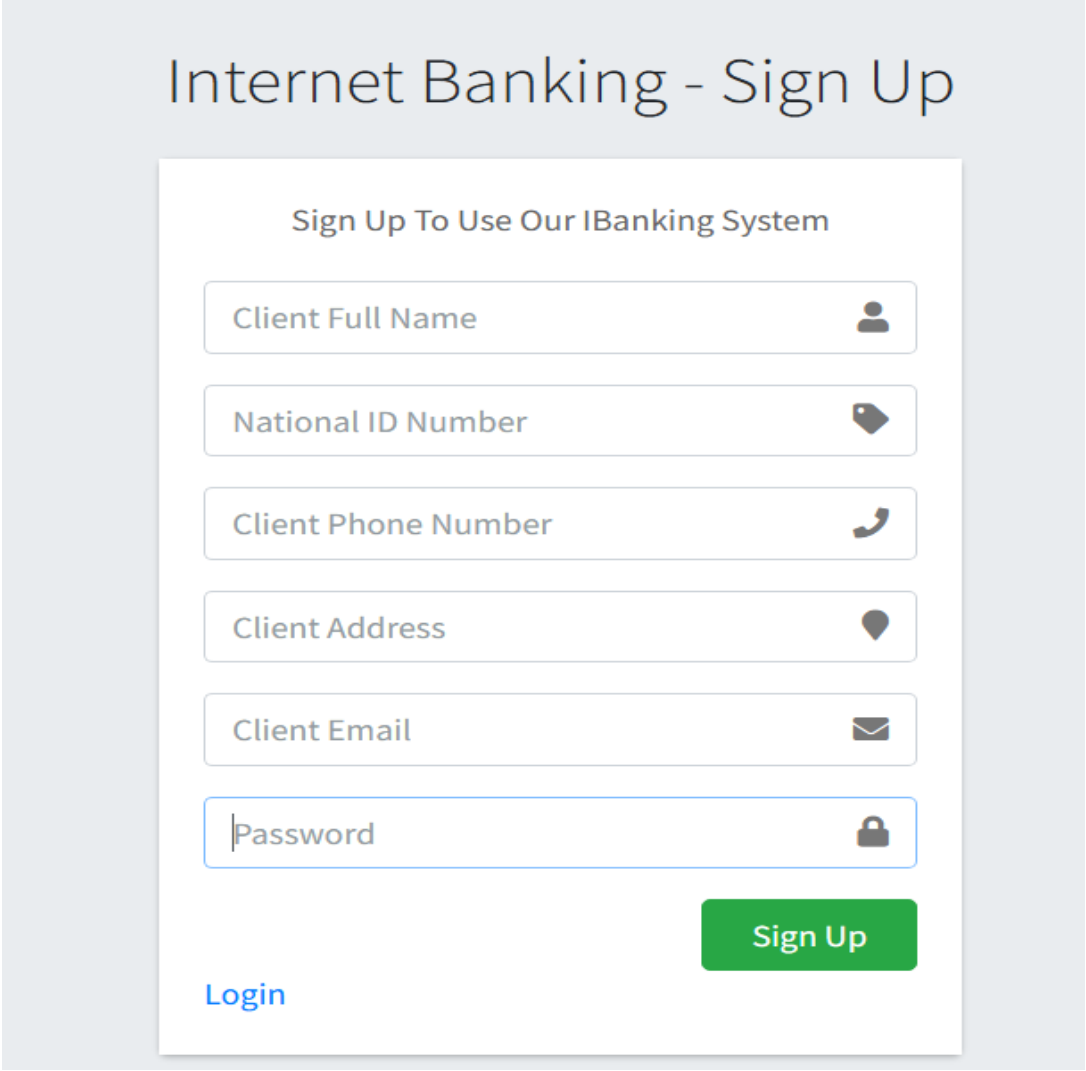
This module allows the Administrator to define and categorize different types of bank accounts offered by the institution. By entering the 'Account Type Name' and the 'Rate of Interest', the Admin can configure the financial rules for each category, which are then used by the system to calculate earnings and manage customer portfolios effectively.

CLIENT ACCOUNT REGISTRATION (STAFF SIDE)



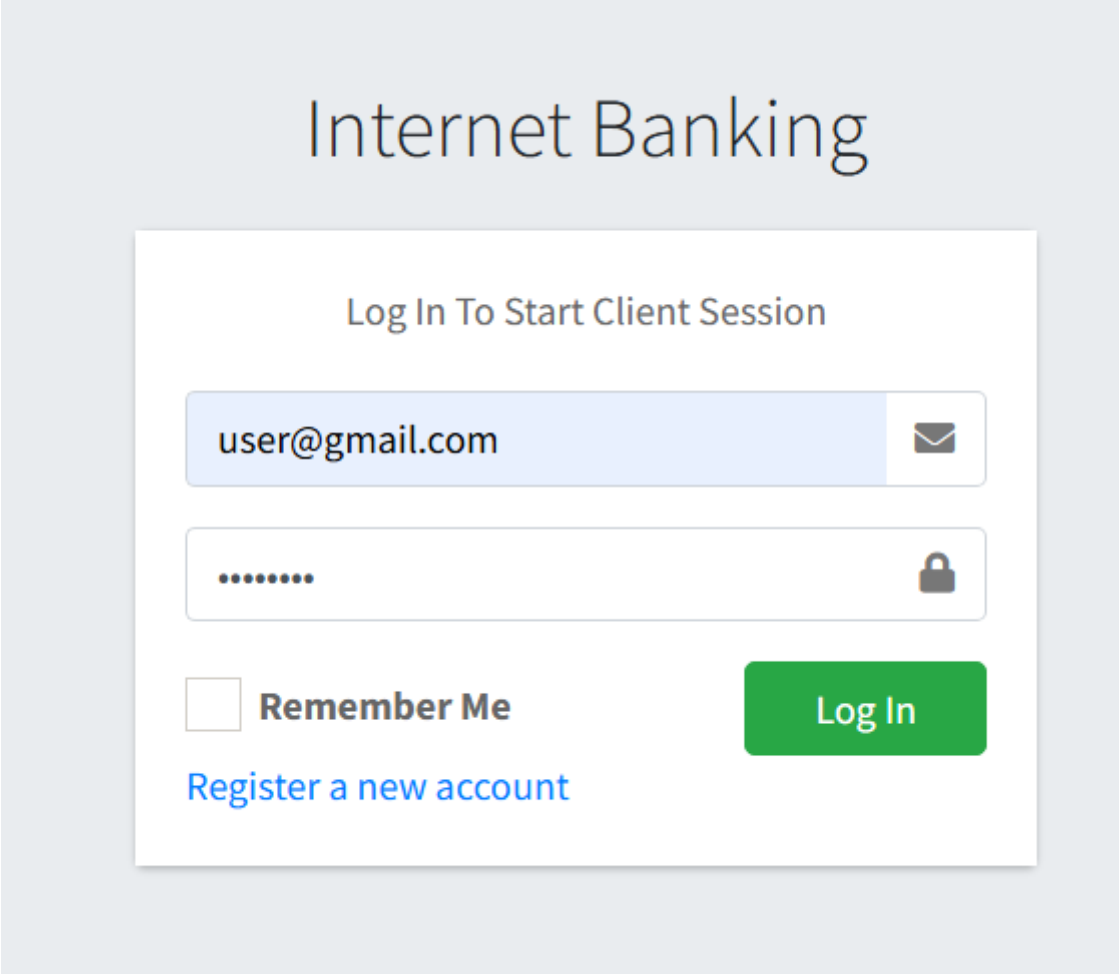
This registration form captures a new customer's personal and contact details to initiate the account opening process. It allows staff to assign specific account types, ensuring a smooth and organized onboarding experience.

SIGN UP (CLIENT SIDE)



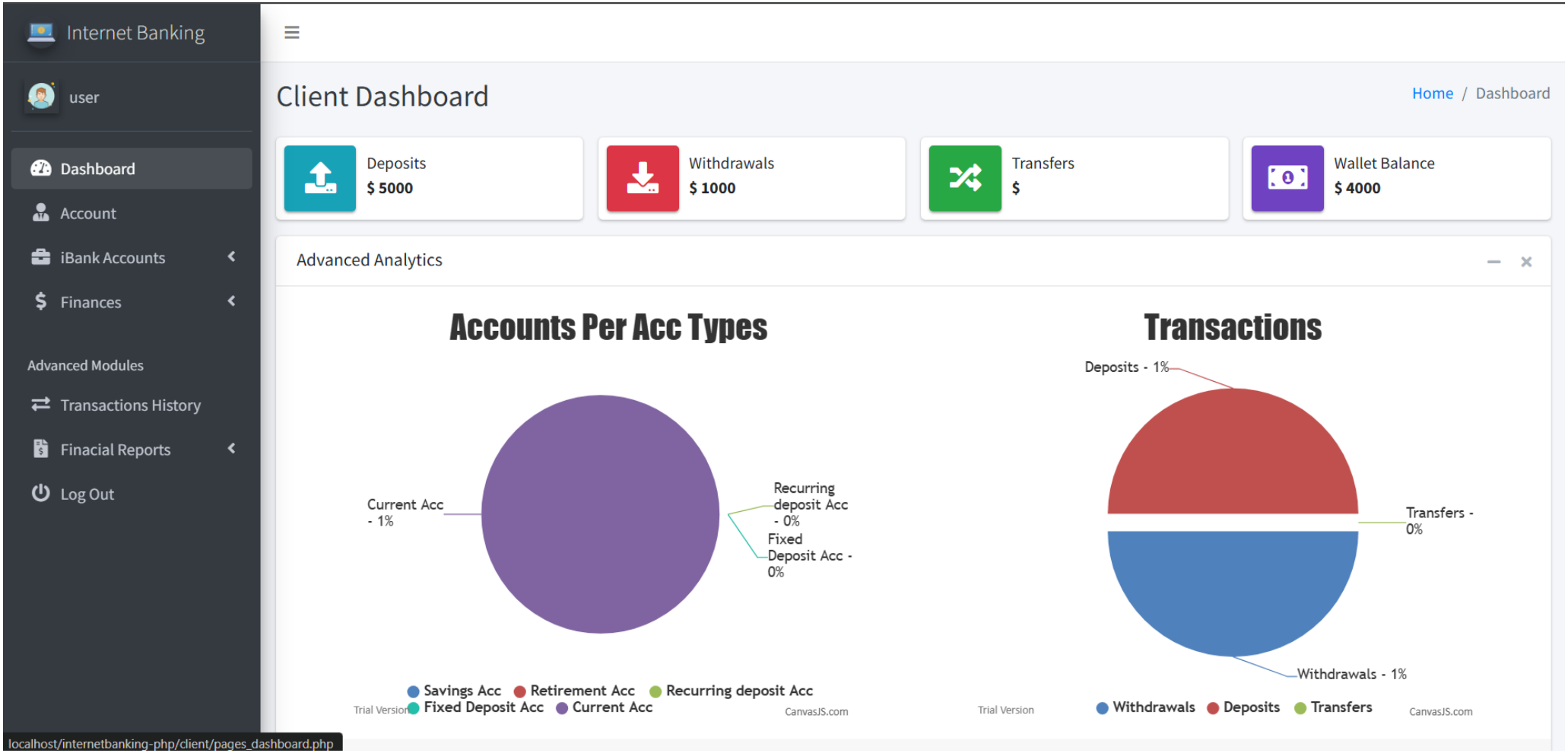
This is the initial onboarding screen for new customers. The registration form captures essential user data, including personal credentials and contact details, which are then securely hashed and stored in the database to create a new user entity.

LOG IN (CLIENT SIDE)



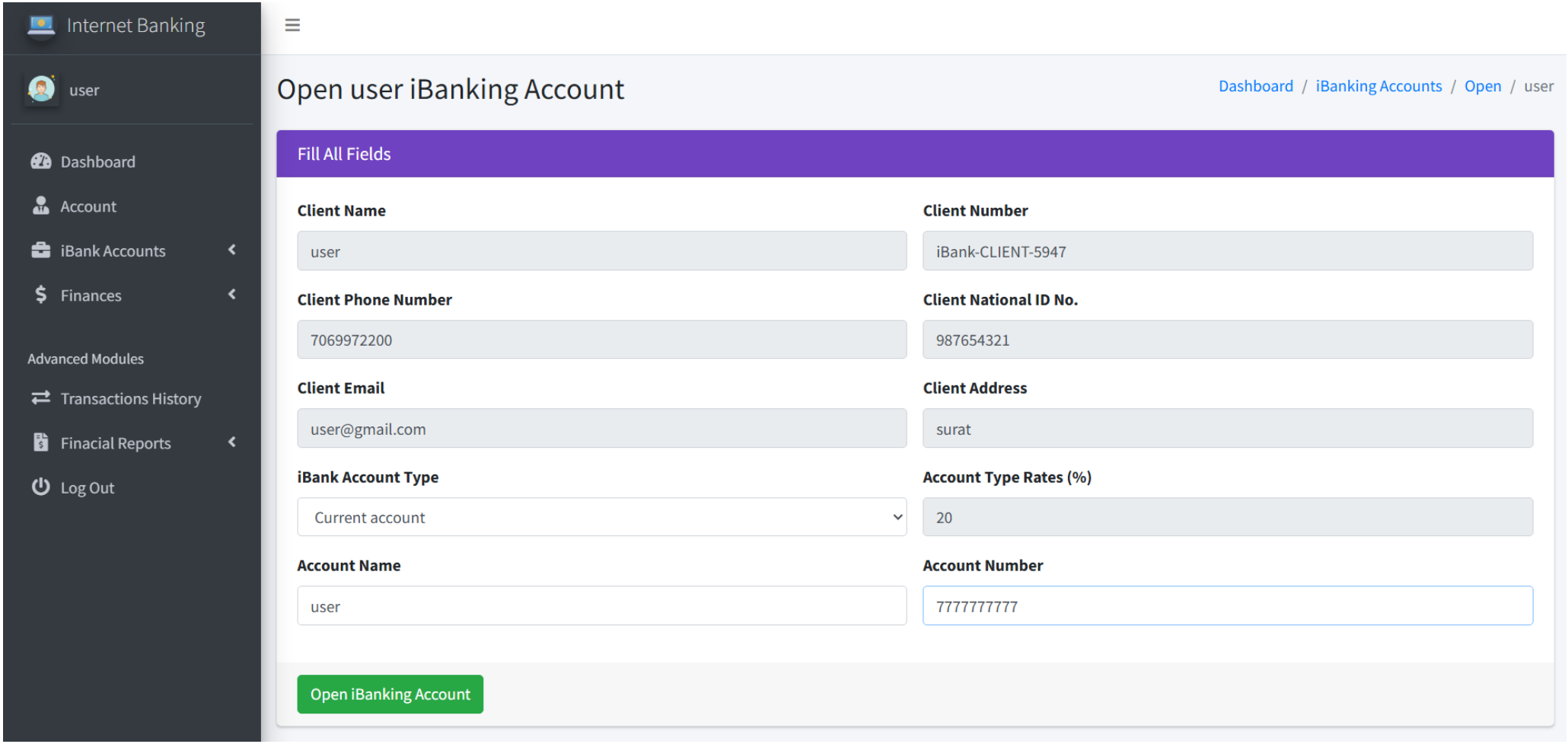
The login portal provides a secure gateway for registered customers. It features session-based authentication to ensure that each user can only access their private financial dashboard, maintaining strict data privacy and preventing unauthorized sessions.

DASHBOARD (CLIENT SIDE)



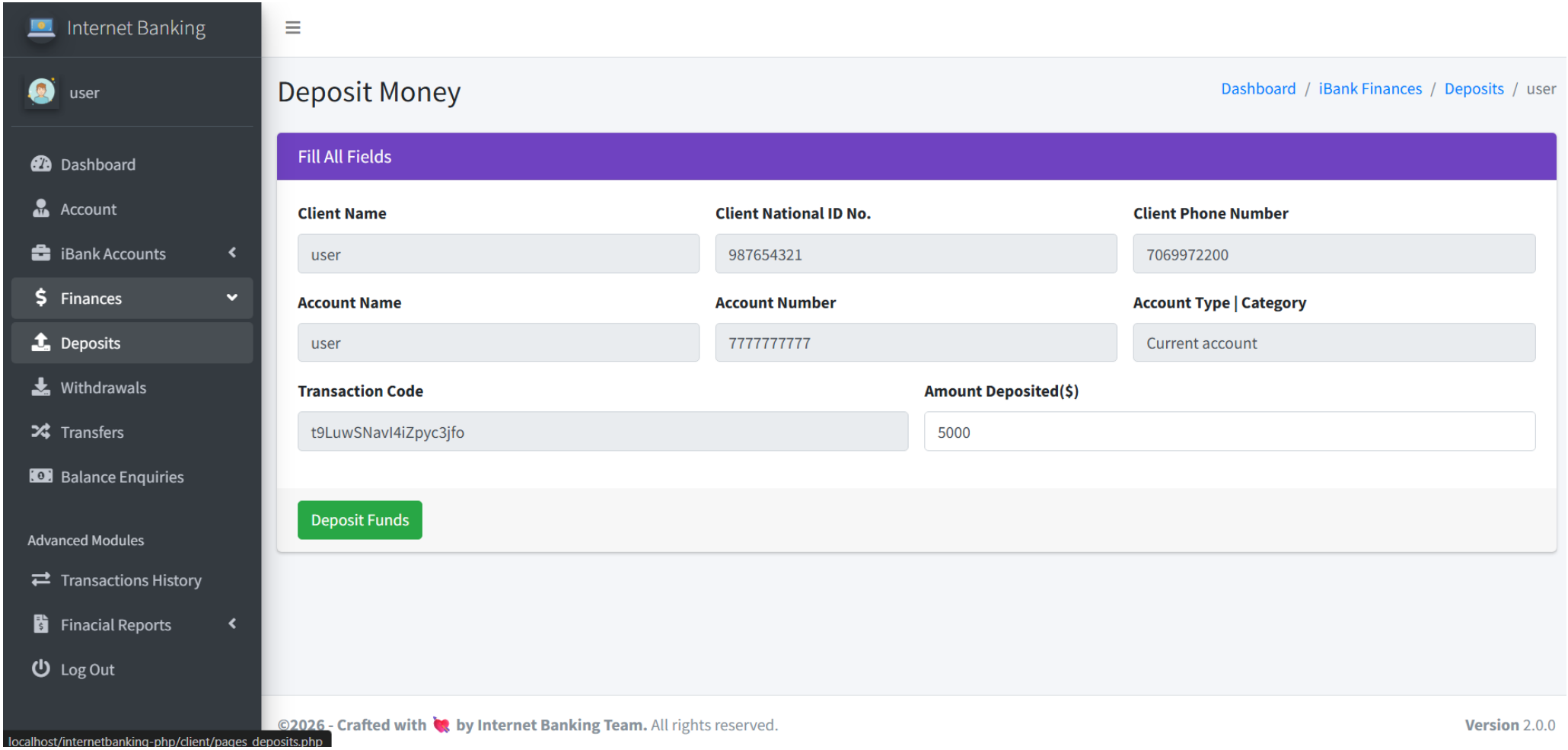
The Dashboard serves as the central hub for the customer. It displays real-time account statistics, including total liquidity and quick-action buttons for transfers or profile updates, providing a seamless user experience (UX) for daily banking tasks.

CREATE ACCOUNT (CLIENT SIDE)



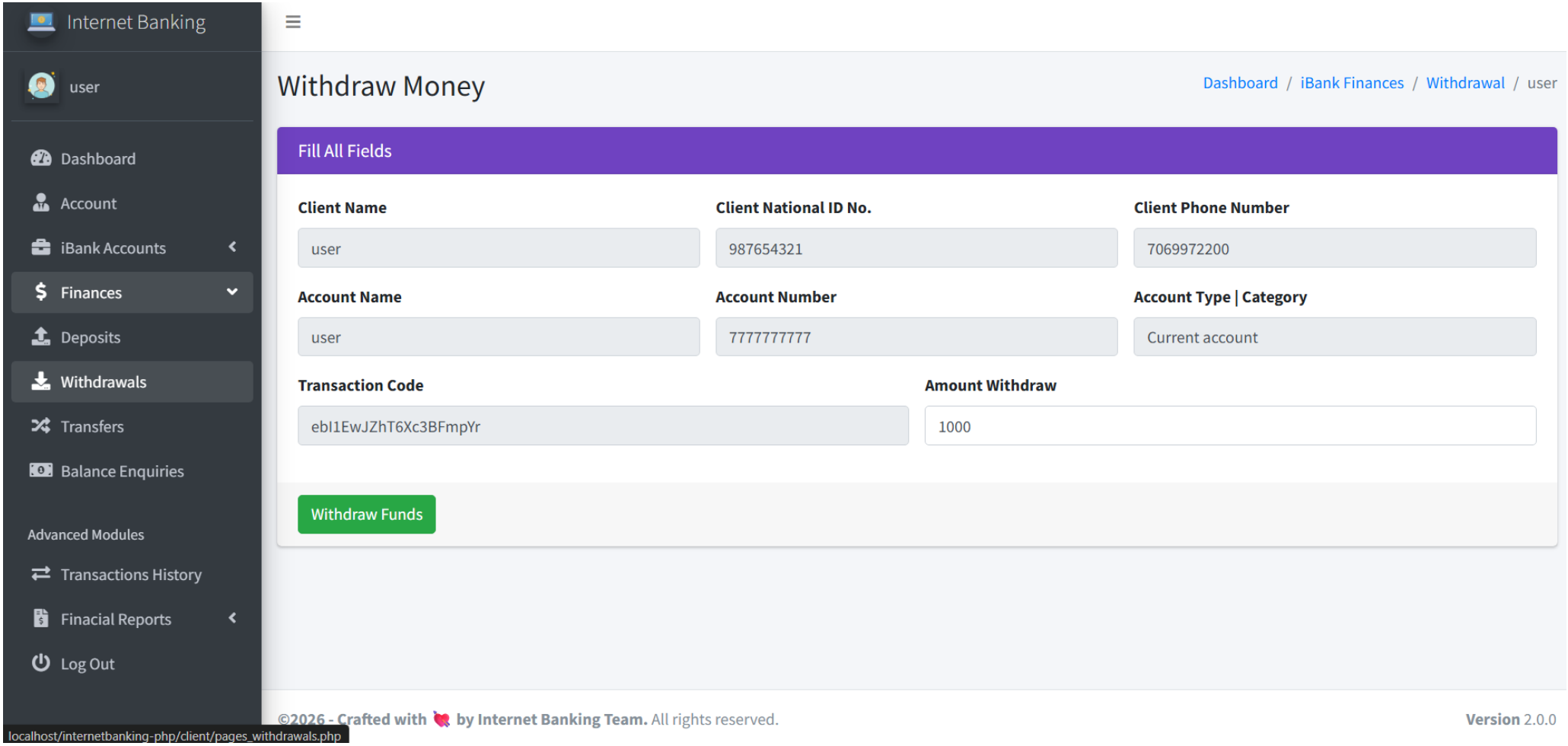
Within the client portal, users can request to open specific types of accounts (Savings/Current). This module bridges the gap between the user's profile and their financial identity by generating a unique 10-digit account number linked to their Client ID.

DEPOSIT MONEY (CLIENT SIDE)



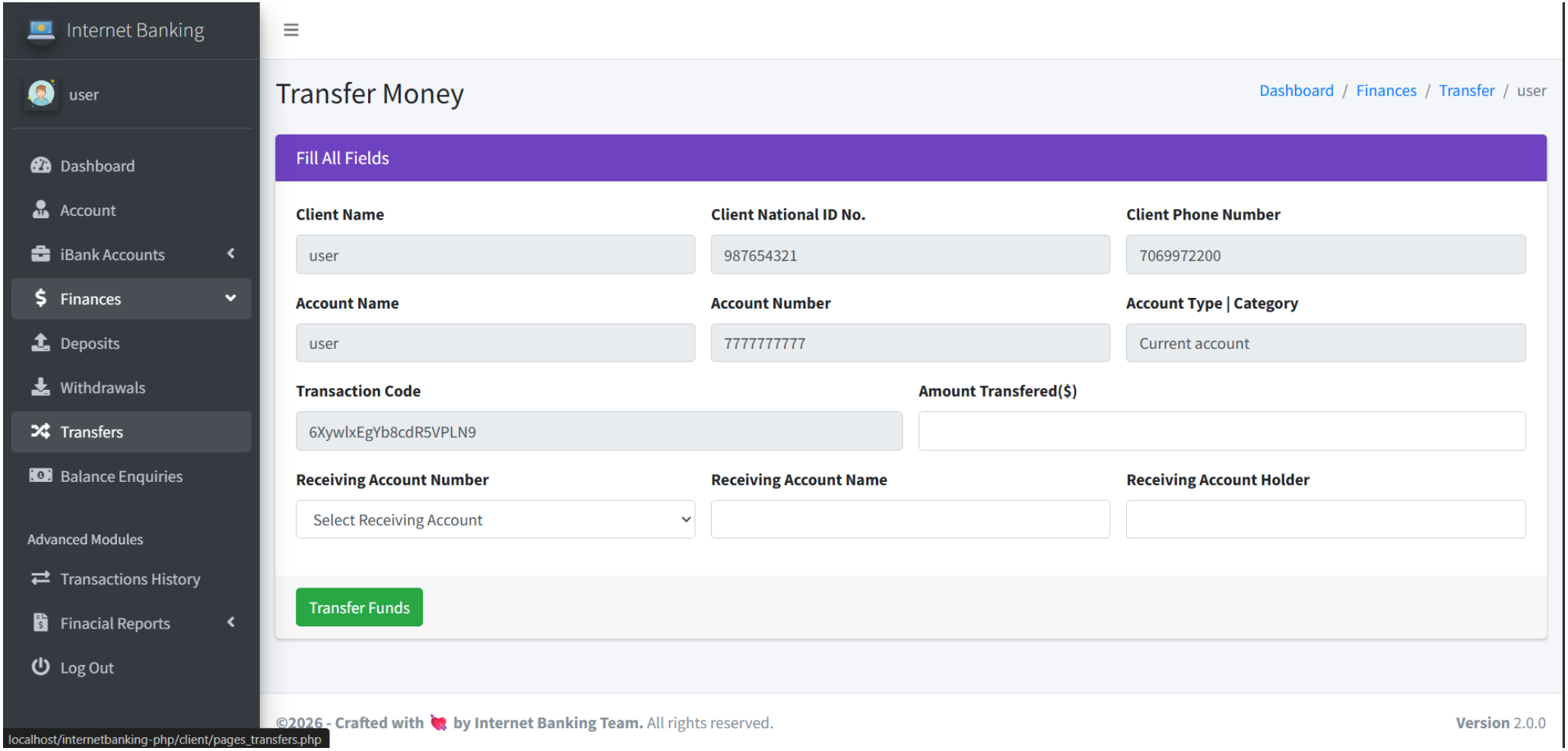
This module is utilized by bank tellers/staff to credit funds into a customer's account. The staff simply verifies the account details and enters the deposit amount, which instantly updates the customer's main ledger balance.

WITHDRAW MONEY (CLIENT SIDE)



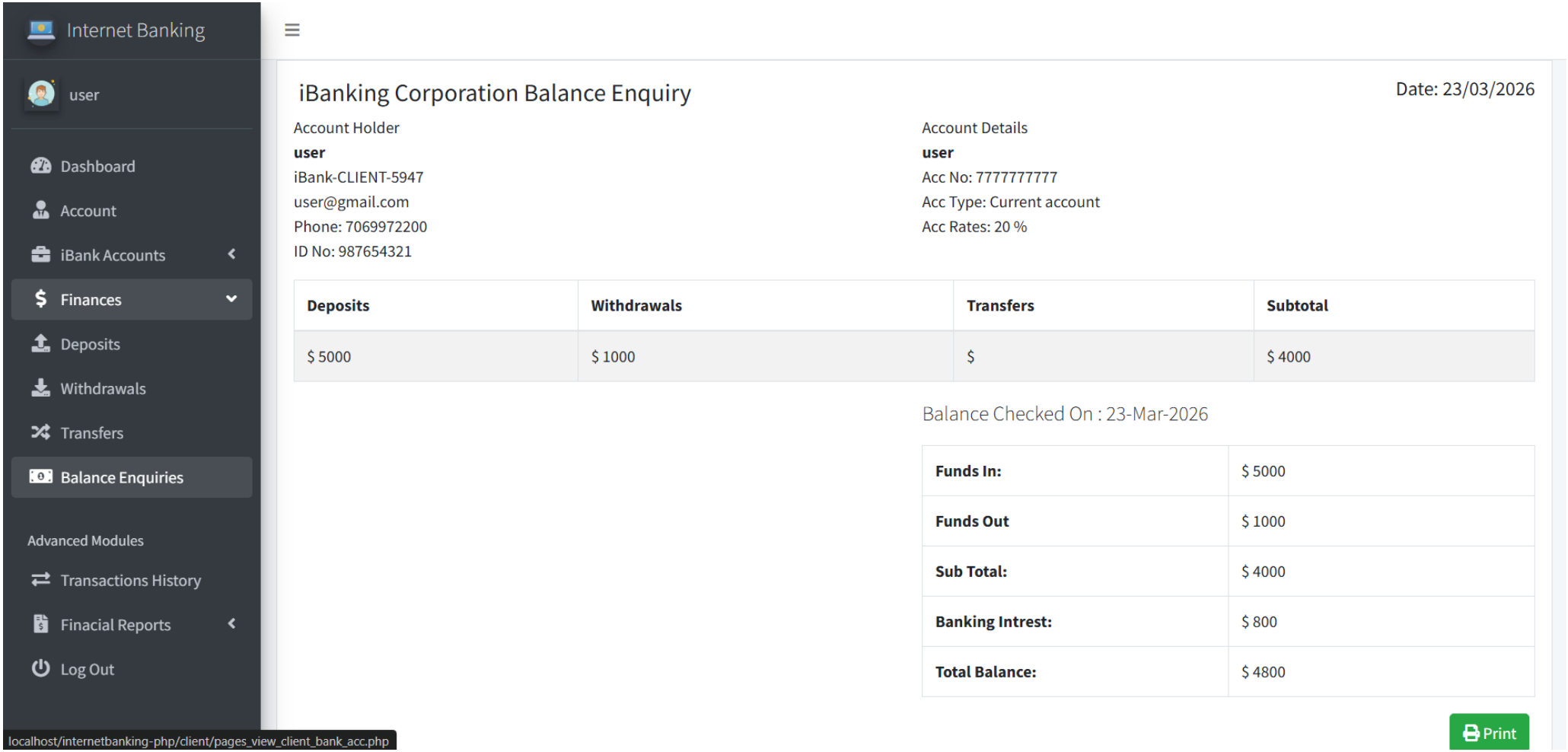
Similar to deposits, this screen handles cash outflows. When a customer requests a physical withdrawal, the staff uses this interface to deduct the specific amount securely from the client's available balance.

TRANSFER MONEY (CLIENT SIDE)



This interface facilitates money transfers between two different accounts within the same bank. It requires the sender's account details, the receiving account number, and the transfer amount, executing the transaction seamlessly in real time.

BALANCE ENQUIRY (CLIENT SIDE)



This module allows customers to perform a real-time fetch of their current ledger balance. It also provides a detailed view of the account's credit and debit history, ensuring that the customer has full transparency over every penny spent or received.

TESTING

Test Case ID	Test Scenario	Input Data	Expected Result	Status
TC-01	Admin Login	Valid Email/Password	Access Admin Dashboard	**Passed**
TC-02	Create New Staff	Name, Email, Dept	New Staff added to Database	**Passed**
TC-03	Open Bank Account	Client Details	Account Number Generated	**Passed**
TC-04	Fund Transfer	Acc No & Amount	Balance deducted/credited	**Passed**
TC-05	System Settings	Update Bank Name	Changes reflect on all pages	**Passed**
TC-06	Security Test	Incorrect Password	Show "Invalid Credentials"	**Passed**

CONCLUSION

The "Internet Banking System" has been successfully developed to bridge the gap between traditional banking and digital convenience. By automating core processes like account management and fund transfers, the system reduces human error and provides 24/7 accessibility. The use of PHP and MySQL ensures a robust backend, while the Bootstrap-based frontend offers a user-friendly experience for both bank staff and customers.

FUTURE SCOPE

While the current system is fully functional, it can be further enhanced in the future by adding:

- **Two-Factor Authentication (2FA):** Implementing OTP via SMS or Email for more security.
- **Mobile Application:** Developing a dedicated Android/iOS app for easier access.
- **Loan Management:** Adding a module where clients can apply for loans online.